

ON THE ASYMPTOTIC NORMALITY OF THE MAXIMUM LIKELIHOOD ESTIMATOR

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SUMMARY

This paper proves the existence of a unique asymptotically normal sequence of solutions to the likelihood equations as a consequence of the Inverse Function Theorem. The approach differs from those in the literature by showing existence, uniqueness, and asymptotic normality at once in the same argument, instead of by showing each aspect separately by differing means. The argument is made without the requirement that the model chosen to construct the likelihood equations be correct.

1. Introduction

The standard argument for asymptotic normality of the maximum likelihood estimator shows that if $\{\hat{\theta}_n\}$ is a consistent sequence of vector parameter estimators that satisfy the corresponding likelihood equations, then a standardized sequence

$$\{n^{1/2}(\hat{\theta}_n - \theta_0)\}$$

converges in distribution to a normally distributed random vector. (See, e.g., Rao 1973, p. 365 and Wilks 1962, pp. 350-362 for the single parameter case and Zacks 1970, pp. 246-247 for the case of a vector parameter.) It is to be emphasized that this argument requires the consistency of the maximum likelihood estimator as a condition for asymptotic normality and that the required consistency must be shown by other means. It is also to be emphasized that the argument does not reveal that the asymptotically normal sequence of solutions to the likelihood equations must exist and be unique.

Classical arguments for consistency in Cramér (1946, p. 500) for the single parameter case and in Chanda (1954) for the vector parameter case show that at least one

consistent sequence of solutions to the likelihood equations must exist. Separate arguments are needed to show that at most one consistent sequence may exist in Huzurbazar (1948) for the single parameter case and in Chanda (1954) for the vector parameter case. However, Foutz (1977) proves the existence of a unique consistent sequence of solutions to the likelihood equations in one argument.

Thus, under appropriate regularity conditions, we are assured of the existence of a unique sequence of solutions to the likelihood equations that is consistent and that is therefore, by the standard argument, asymptotically normal; but our route to this conclusion has been an indirect one requiring separate arguments along the way. The purpose of this paper is to show directly, in a single argument, that there must exist a unique asymptotically normal sequence of solutions to the likelihood equations.

The asymptotic normality of the maximum likelihood estimator has also been examined in cases where the model chosen for constructing the estimator may be incorrect. The model is said to be incorrect when the distribution of the observations is not contained in the family of distributions from which the likelihood function is constructed. In particular, Huber (1965) argues asymptotic normality of the maximum likelihood estimator without the requirement that the model need be correct; however, the argument is made "Assuming that consistency ... has already been proved by some other means ..." Other treatments of the maximum likelihood estimator when the model is incorrect are found in Pfanzagl (1969, 1971, 1973); Michel and Pfanzagl (1971); Perlman (1972); and Berk (1972). The argument of this paper is made without the requirement that the model chosen to construct the likelihood equations be correct.

2. Preliminaries

Let $X, X_1, X_2, \dots$ be a sequence of independent identically distributed random variables defined on a probability space $(X, \mathcal{B}, P)$. All expectations are understood to be with respect to the probability P , and sequences of statements $A_1(X_1), A_2(X_1, X_2), \dots$ are said to hold with probability going to one if $\lim_{n \rightarrow \infty} P[A_n(X_1, X_2, \dots, X_n)] = 1$.

Let Θ be a subset of Euclidean r space, E^r , and take $\{P_\theta, \Theta\}$ to be a family of distinct probabilities on $(X, \mathcal{B})$. It is not necessarily assumed that the family contains P . For $\{f(x, \theta), \Theta\}$ a corresponding family of densities with respect to a σ -finite measure on $(X, \mathcal{B})$, the following conditions are required:

A₁. For almost all x , the second order partial derivatives, $\partial^2 \log f(x, \theta) / \partial \theta_i \partial \theta_j$, $i, j = 1, 2, \dots, r$ exist and are continuous on Θ .

A₂. There exists a θ_0 in Θ for which $E \partial \log f(X, \theta) / \partial \theta_j |_{\theta_0} = 0$ for $j = 1, 2, \dots, r$.

A₃. There is a neighbourhood about θ_0 on which the convergence in probability

of $n^{-1} \sum_{k=1}^n \partial^2 \log f(X_k, \underline{\theta}) / \partial \theta_i \partial \theta_j$ to $E \partial^2 \log f(X, \underline{\theta}) / \partial \theta_i \partial \theta_j$ is uniform for $i = 1, 2, \dots, r$ and $j = 1, 2, \dots, r$.

A₄. The r dimensional square matrix $\underline{g}(\underline{\theta})$ with ij^{th} element $E \partial^2 \log f(X, \underline{\theta}) / \partial \theta_i \partial \theta_j$ is continuous and nonsingular at $\underline{\theta} = \underline{\theta}_0$. Thus, the inverse matrix $\underline{g}(\underline{\theta}_0)^{-1}$ is well defined.

Remark 2.1

When the model is correct and $f(x, \underline{\theta}_0)$ is a density for the true probability P , Wald (1949, Lemma 1) shows that $E \log f(X, \underline{\theta}) < E \log f(X, \underline{\theta}_0)$ for $\underline{\theta} \neq \underline{\theta}_0$. Thus, in this case, A₂ holds with $\underline{\theta}_0$ corresponding to the usual notion of the true value of $\underline{\theta}$. When the model is incorrect, the quantity $E \log f(X, \underline{\theta})$ may be thought of as a measure of the "similarity" between the true probability P and the probability $P_{\underline{\theta}}$ in the model $\{P_{\underline{\theta}}, \Theta\}$. In fact, $E \log f(x, \underline{\theta})$ is a function of Kullback's directed divergence between probability distributions (cf. Kullback 1959, p. 7). Accordingly, since $\underline{\theta}_0$ maximizes $E \log f(X, \underline{\theta})$ over Θ , we may think of $P_{\underline{\theta}_0}$ as the modelled probability that is "most similar" to P .

The log likelihood function is constructed from the model to be

$$L_n(\underline{\theta}) = \sum_{i=1}^n \log f(X_i, \underline{\theta}).$$

For the vector of derivatives of $\log f(x, \underline{\theta})$, write

$$\psi(x, \underline{\theta}) = (\partial \log f(x, \underline{\theta}) / \partial \theta_1, \partial \log f(x, \underline{\theta}) / \partial \theta_2, \dots, \partial \log f(x, \underline{\theta}) / \partial \theta_r)',$$

and note that the random functions

$$\partial L_n(\underline{\theta}) / \partial \underline{\theta} = \sum_{i=1}^n \psi(X_i, \underline{\theta})$$

and

$$\underline{S}_n(\underline{\theta}) = n^{-1} \partial L_n(\underline{\theta}) / \partial \underline{\theta}$$

take Θ into E^r . The maximum likelihood estimator of $\underline{\theta}$ is often taken to be a solution to

the vector likelihood equations $\underline{S}_n(\underline{\theta}) = \underline{0}$.

From another point of view, the maximum likelihood estimator is the value of $\underline{Q}$ of the random inverse function $\underline{S}_n^{-1}: \underline{S}_n(\Theta) \rightarrow \Theta$. The argument of this paper is made from this point of view. The principal tool of the argument is the Inverse Function Theorem, a nonstandard statement of which is given here. For the statement take the norm $\|\underline{C}\|$ of an r dimensional square matrix $\underline{C}$ to be the lub of all numbers $|\underline{C}\underline{x}|$, where $\underline{x}$ ranges over all vectors in E^r with $|\underline{x}| \leq 1$.

The Inverse Function Theorem.

Suppose $\underline{f}$ is a mapping from an open set Θ in E^r into E^r , the partial derivatives of $\underline{f}$ exist and are continuous on Θ , and the matrix of derivatives $\underline{f}'(\underline{\theta}_0)$ has inverse $\underline{f}'(\underline{\theta}_0)^{-1}$ for some $\underline{\theta}_0$ in Θ . Write $\lambda = 1/(4\|\underline{f}'(\underline{\theta}_0)^{-1}\|)$. Use the continuity of the elements of $\underline{f}'(\underline{\theta}_0)$ to fix a neighbourhood $N_\delta(\underline{\theta}_0)$ of $\underline{\theta}_0$ with sufficiently small radius δ to ensure $\|\underline{f}'(\underline{\theta}) - \underline{f}'(\underline{\theta}_0)\| < 2\lambda$, whenever $\underline{\theta} \in N_\delta(\underline{\theta}_0)$. Then $\underline{f}$ is one-to-one on $N_\delta(\underline{\theta}_0)$, and the image set $\underline{f}(N_\delta(\underline{\theta}_0))$ contains the open neighbourhood with radius $\lambda\delta$ about $\underline{f}(\underline{\theta}_0)$.

This form of the theorem is proved in Rudin (1964, p. 193).

The following property of the matrix norm $\|\underline{C}\|$ is proved in Rudin (1964, p. 186).

Lemma 2.1.

If the r dimensional square matrix $\underline{C}$ is invertible with $\|\underline{C}^{-1}\| = 1/\alpha$, then the second r dimensional square matrix $\underline{D}$ is invertible if $\|\underline{C} - \underline{D}\| < \alpha$.

3. Existence, Uniqueness, and Asymptotic Normality

The r dimensional square matrix $\underline{I}(\underline{\theta})$ with ij^{th} element $E(\partial \log f(X, \underline{\theta}) / \partial \theta_i) (\partial \log f(X, \underline{\theta}) / \partial \theta_j)$, $i, j = 1, 2, \dots, r$, is required for the statement of the main result. The result gives the existence of a unique, asymptotically normal sequence of solutions to the likelihood equations, $\underline{S}_n(\underline{\theta}) = \underline{0}$.

Theorem 3.1

There exists a sequence of estimators, $\{\hat{\underline{\theta}}_n\}$, for which

$$(3.1) \quad \underline{S}_n(\hat{\underline{\theta}}_n) = \underline{0}$$

with probability going to one and for which the convergence in distribution

$$(3.2) \quad n^{1/2}(\hat{\theta}_n - \theta_0) \rightarrow \tilde{G}$$

holds for $\tilde{G}$ an r variate normal random vector with mean vector Q and covariance matrix $\underline{g}(\theta_0)^{-1} \underline{I}(\theta_0) \underline{g}(\theta_0)^{-1}$. The sequence is unique, for if $\{\hat{\theta}_n\}$ also satisfies (3.1) and satisfies (3.2) for some normally distributed $\tilde{G}$, then $\bar{\theta}_n = \hat{\theta}_n$ with probability going to one.

Proof.

With probability going to one the conditions of the Inverse Function Theorem hold for the random function $\underline{S}_n: \Theta \rightarrow E^r$. This is because $\underline{S}_n$ is continuously differentiable, by A_1 , and because the matrix $\underline{S}_n'(\theta_0)$ has inverse $\underline{S}_n'(\theta_0)^{-1}$ with probability going to one, by A_4 , A_3 and Lemma 2.1.

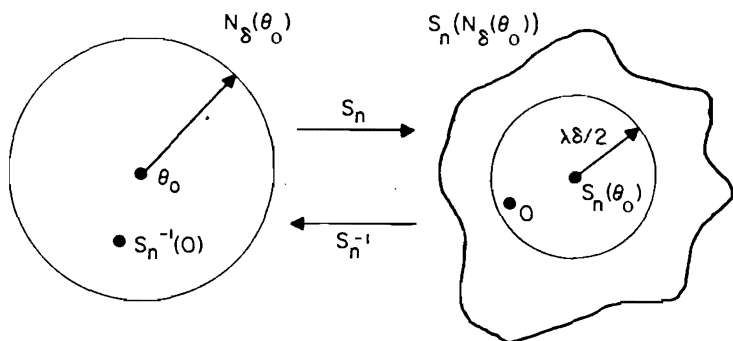
Write $\lambda = 1/(4\|\underline{g}(\theta_0)^{-1}\|)$. By the continuity of $\underline{g}(\theta)$ at $\theta = \theta_0$ there exists an open neighbourhood, $N_\delta(\theta_0)$, with radius δ about θ_0 such that $\|\underline{g}(\theta) - \underline{g}(\theta_0)\| < \lambda/2$ whenever $\theta \in N_\delta(\theta_0)$ and on which the uniform convergence of $\underline{S}_n'(\theta)$ to $\underline{g}(\theta)$ holds according to A_3 . Write $\lambda_n = 1/(4\|\underline{S}_n'(\theta_0)^{-1}\|)$ whenever $\underline{S}_n'(\theta)$ is invertible, and note that $\lambda_n \rightarrow \lambda$ in probability. Finally, use the uniform convergence in probability of $\underline{S}_n'(\theta)$ to $\underline{g}(\theta)$ for $\theta \in N_\delta(\theta_0)$ to argue

$$\begin{aligned} \|\underline{S}_n'(\theta) - \underline{S}_n'(\theta_0)\| &\leq \|\underline{S}_n'(\theta) - \underline{g}(\theta)\| + \|\underline{g}(\theta_0) - \underline{S}_n'(\theta_0)\| + \|\underline{g}(\theta) - \underline{g}(\theta_0)\| \\ &< \lambda \\ &< 2\lambda_n \end{aligned}$$

with probability going to one as n increases.

We now see that the Inverse Function Theorem ensures with probability going to one that $\underline{S}_n$ is one-to-one on $N_\delta(\theta_0)$ and that the image set $\underline{S}_n(N_\delta(\theta_0))$ contains the open neighbourhood of radius $\lambda_n \delta$ about $\underline{S}_n(\theta_0)$. Since $\lambda_n \delta \rightarrow \lambda \delta$ in probability, $\underline{S}_n(N_\delta(\theta_0))$ also contains the neighbourhood of radius $\lambda \delta/2$ about $\underline{S}_n(\theta_0)$ with probability going to one. This neighborhood contains the zero vector with probability going to one since A_2 and the Law of Large Numbers ensure that $\|\underline{S}_n(\theta_0) - Q\|$ converges in probability to 0.

Combined, these conclusions show the construction of the figure to hold with probability going to one.

THE FUNCTIONS S_n AND S_n^{-1}

Under the conditions of the figure, the existence of 0 in $S_n(N_{\delta}(\theta_0))$ ensures the corresponding existence of a solution to the likelihood equations, $S_n^{-1}(0)$, in $N_{\delta}(\theta_0)$; in addition, the one-to-oneness of S_n on $N_{\delta}(\theta_0)$ ensures $S_n^{-1}(0)$ to be the likelihood equations' unique solution in $N_{\delta}(\theta_0)$.

To speak precisely of the sequence of estimators $\{S_n^{-1}(0)\}$, define $S_n^{-1}(0)$ to be the unique solution to the likelihood equations in $N_{\delta}(\theta_0)$ when the conditions of the figure are met, that is, with probability going to one; and define $S_n^{-1}(0)$ arbitrarily otherwise. By this construction, $S_n^{-1}(0)$ exists for each n ; and with probability going to one, it is the unique solution to the likelihood equations in $N_{\delta}(\theta_0)$. This uniqueness of $S_n^{-1}(0)$ in $N_{\delta}(\theta_0)$ requires the members of another sequence $\{\bar{\theta}_n\}$ of solutions to the likelihood equations to lie outside of $N_{\delta}(\theta_0)$ with probability going to one so that the standardized sequence $\{n^{1/2}(\bar{\theta}_n - \theta_0)\}$ cannot be asymptotically normal.

To argue asymptotic normality of the sequence $\{n^{1/2}(S_n^{-1}(0) - \theta_0)\}$, select any bounded open set B in E^r , and consider the probability $P[n^{1/2}(S_n^{-1}(0) - \theta_0) \in B]$. With $H_n = \{\theta_0 + h : n^{1/2}h \in B\}$, we have the equality

$$(3.3) \quad P[n^{1/2}(S_n^{-1}(0) - \theta_0) \in B] = P[S_n^{-1}(0) \in H_n].$$

Since B is a bounded set, it follows that H_n shrinks to the singleton set $\{\underline{\theta}_0\}$ as n increases. Thus for all sufficiently large n , H_n is a subset of $N_\delta(\underline{\theta}_0)$, and $\underline{S}_n$ is thereby one-to-one on H_n when the conditions of the figure hold. Since these conditions hold with probability going to one, we have from (3.3) that

$$(3.4) \quad \lim_{n \rightarrow \infty} P[n^{1/2}(\underline{S}_n^{-1}(\underline{Q}) - \underline{\theta}_0) \in B] - P[\underline{Q} \in \underline{S}_n(H_n)] = 0,$$

where $\underline{S}_n(H_n) = \{\underline{S}_n(\underline{\theta}_0 + \underline{h}) : n^{1/2}\underline{h} \in B\}$.

By the definition of the total derivative $\underline{S}_n'(\underline{\theta}_0)$, we may write $\underline{S}_n(\underline{\theta}_0 + \underline{h})$ as $\underline{S}_n(\underline{\theta}_0) + \underline{S}_n'(\underline{\theta}_0)\underline{h} + \underline{r}_n(\underline{h})$, where the function $\underline{r}_n(\underline{h}) = \underline{S}_n(\underline{\theta}_0 + \underline{h}) - \underline{S}_n(\underline{\theta}_0) - \underline{S}_n'(\underline{\theta}_0)\underline{h}$ satisfies

$$(3.5) \quad \lim_{|\underline{h}| \rightarrow 0} |\underline{r}_n(\underline{h})| / |\underline{h}| = 0.$$

The expression (3.4) may be written accordingly as

$$(3.6) \quad \lim_{n \rightarrow \infty} P[(n^{1/2}(\underline{S}_n^{-1}(\underline{Q}) - \underline{\theta}_0) \in B] - P[\underline{Q} \in \{\underline{S}_n(\underline{\theta}_0) + \underline{S}_n'(\underline{\theta}_0)\underline{h} + \underline{r}_n(\underline{h}) : n^{1/2}\underline{h} \in B\}] = 0.$$

With $B_n = \{n^{1/2}\underline{h} + n^{1/2}\underline{S}_n'(\underline{\theta}_0)^{-1}\underline{r}_n(\underline{h}) : n^{1/2}\underline{h} \in B\}$, expression (3.6) becomes

$$(3.7) \quad \lim_{n \rightarrow \infty} P[n^{1/2}(\underline{S}_n^{-1}(\underline{Q}) - \underline{\theta}_0) \in B] - P[-n^{1/2}\underline{S}_n'(\underline{\theta}_0)^{-1}\underline{r}_n(\underline{\theta}_0) \in B_n] = 0.$$

The set B_n is the image of the bounded open set B under the random mapping $\underline{T}_n(\underline{b}) = \underline{b} + n^{1/2}\underline{S}_n'(\underline{\theta}_0)^{-1}\underline{r}_n(n^{-1/2}\underline{b})$. The conditions of the Inverse Function Theorem hold for $\underline{T}_n$ with probability going to one, since the total derivative of $\underline{T}_n$ at $\underline{b}$ exists as

$$\begin{aligned} \underline{T}_n'(\underline{b}) &= \underline{I} + n^{1/2}\underline{S}_n'(\underline{\theta}_0)^{-1}[\underline{S}_n'(\underline{\theta}_0 + n^{-1/2}\underline{b}) - \underline{S}_n'(\underline{\theta}_0)]n^{-1/2} \\ &= \underline{S}_n'(\underline{\theta}_0)^{-1}\underline{S}_n'(\underline{\theta}_0 + n^{-1/2}\underline{b}) \end{aligned}$$

whenever $\underline{S}_n'(\underline{\theta}_0)$ is invertible, that is, with probability going to one. Also, since B is bounded, $\underline{T}_n'(\underline{b})$ converges in probability to the r dimensional identity matrix $\underline{I}$ uniformly for $\underline{b}$ in B ; so it follows from Lemma 2.1 that $\underline{T}_n'(\underline{b})^{-1}$ exists for each $\underline{b}$ in B

with probability going to one.

Write $\gamma_n = 1/(4 \|\mathcal{T}_n'(\underline{b})^{-1}\|)$ whenever $\mathcal{T}_n'(\underline{b})$ is invertible, and note that γ_n converges in probability to $1/4$, uniformly for $\underline{b}$ in B .

Pick any positive ε , and let $N_\varepsilon(\underline{b})$ be the neighbourhood of radius ε about the point $\underline{b}$ in B . By the uniform convergence in probability of $\mathcal{T}_n'(\underline{b})$ to $\mathcal{I}$, we have

$$\|\mathcal{T}_n'(\underline{b}^*) - \mathcal{T}_n'(\underline{b}^{**})\| \leq \|\mathcal{T}_n'(\underline{b}^*) - \mathcal{I}\| + \|\mathcal{I} - \mathcal{T}_n'(\underline{b}^{**})\| < 2\gamma_n$$

simultaneously for every $\underline{b}^*$ and $\underline{b}^{**}$ in B , with probability going to one. It follows with probability going to one from the Inverse Function Theorem that simultaneously for each $\underline{b}$ in B the image set $\mathcal{T}_n(N_\varepsilon(\underline{b}))$ contains the corresponding open neighbourhood of radius $\varepsilon\gamma_n$ about $\mathcal{T}_n(\underline{b})$. Since $\gamma_n \rightarrow 1/4$ in probability uniformly for $\underline{b}$ in B , it also follows that simultaneously for each $\underline{b}$ in B , $\mathcal{T}_n(N_\varepsilon(\underline{b}))$ contains the neighbourhood $N_{\varepsilon/8}(\mathcal{T}_n(\underline{b}))$ with probability going to one.

The set $B_\varepsilon^- = \{\underline{b} \in B: N_\varepsilon(\underline{b}) \subseteq B\}$ is a subset of $\mathcal{T}_n^+(B) = B_n$ with probability going to one, and B_n is in turn a subset of $B_\varepsilon^+ = \bigcup_{\underline{b} \in B} N_\varepsilon(\underline{b})$ with probability going to one. To see this, let $\tilde{r}_{i,n}(\underline{h})$ be the i th component of the vector function $\underline{r}_n(\underline{h})$ satisfying (3.5); and use the argument of Rudin (1964, p. 192) to write

$$\tilde{r}_{i,n}(\underline{h}) = \left(n^{-1} \sum_{k=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_1} \log f(X_k, \underline{\theta}) \Big|_{\underline{\theta}_0 + \underline{\omega}_{j1}} - n^{-1} \sum_{k=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_1} \log f(X_k, \underline{\theta}) \Big|_{\underline{\theta}_0}, \dots, \right.$$

(3.8)

$$\left. n^{-1} \sum_{k=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_r} \log f(X_k, \underline{\theta}) \Big|_{\underline{\theta}_0 + \underline{\omega}_{ir}} - n^{-1} \sum_{k=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_r} \log f(X_k, \underline{\theta}) \Big|_{\underline{\theta}_0} \right) \underline{h}$$

for $\underline{\omega}_{ij}$ satisfying $|\underline{\omega}_{ij}| \leq |\underline{h}|$, for $j = 1, 2, \dots, r$. From (3.8) a bound on $|\tilde{r}_n(\underline{h})|$ is

$$|\tilde{r}_n(\underline{h})| \leq r^2 \max_{i,j=1, \dots, r} \left| n^{-1} \sum_{k=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(X_k, \underline{\theta}) \Big|_{\underline{\theta}_0 + \underline{\omega}_{ij}} - \right.$$

(3.9)

$$\left. n^{-1} \sum_{k=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(X_k, \underline{\theta}) \Big|_{\underline{\theta}_0} \right| |\underline{h}|.$$

Fix $\beta > 0$. Conditions Λ_3 and Λ_4 may be used to replace the bound in (3.9), with probability going to one, by

$$(3.10) \quad |\tau_n(\underline{h})| \leq \beta |\underline{h}|$$

whenever $|\underline{h}|$ is sufficiently small. If we now write $|\underline{b} - \tau_n(\underline{b})| = n^{1/2} |\underline{S}_n'(\underline{\varrho}_0)^{-1} \tau_n(n^{-1/2} \underline{b})|$, then we may conclude from (3.10) that the right side is smaller than $\epsilon/8$ simultaneously for all $\underline{b} \in B$, with probability going to one. Thus every neighbourhood $N_{\epsilon/8}(\tau_n(\underline{b}))$ contains $\underline{b}$ with probability going to one. But we have shown that simultaneously for every $\underline{b}$ in B , $\tau_n(N_\epsilon(\underline{b})) \supseteq N_{\epsilon/8}(\tau_n(\underline{b}))$ with probability going to one, so the results together imply that simultaneously every $\underline{b}$ in B is in $\tau_n(N_\epsilon(\underline{b}))$ with probability going to one. This shows that $B_\epsilon^- \subset B_n \subset B_\epsilon^+$ with probability going to one.

Since $B_\epsilon^- \subset B_n \subset B_\epsilon^+$ with probability going to one, expression (3.7) may be used to write

$$(3.11) \quad \lim_{n \rightarrow \infty} P[n^{1/2}(\underline{S}_n^{-1}(\underline{Q}) - \underline{\varrho}_0) \in B] - P[-n^{1/2} \underline{S}_n'(\underline{\varrho}_0)^{-1} \underline{S}_n(\underline{\varrho}_0) \in B_\epsilon^+] \leq 0 \\ \leq \lim_{n \rightarrow \infty} P[n^{1/2}(\underline{S}_n^{-1}(\underline{Q}) - \underline{\varrho}_0) \in B] - P[-n^{1/2} \underline{S}_n'(\underline{\varrho}_0)^{-1} \underline{S}_n(\underline{\varrho}_0) \in B_\epsilon^-].$$

The multivariate central limit theorem applied to the sample mean vector $-\underline{S}_n(\underline{\varrho}_0) = -n^{-1} \sum_{i=1}^n \psi(X_i, \underline{\varrho}_0)$ ensures that $-n^{1/2} \underline{S}_n(\underline{\varrho}_0)$ is asymptotically normally distributed with mean vector $\underline{Q}$ and covariance matrix $\underline{I}(\underline{\varrho}_0)$. Since $\underline{S}_n'(\underline{\varrho}_0)^{-1}$ converges in probability to $\underline{g}(\underline{\varrho}_0)^{-1}$, it follows that $-n^{1/2} \underline{S}_n'(\underline{\varrho}_0)^{-1} \underline{S}_n(\underline{\varrho}_0)$ converges in distribution to $\underline{G}$, for $\underline{G}$ an r variate normally distributed vector with mean vector $\underline{Q}$ and covariance matrix $\underline{g}(\underline{\varrho}_0)^{-1} \underline{I}(\underline{\varrho}_0) \underline{g}(\underline{\varrho}_0)^{-1}$. Therefore, (3.11) may be restated as

$$(3.12) \quad P[\underline{G} \in B_\epsilon^-] \leq \lim_{n \rightarrow \infty} P[n^{1/2}(\underline{S}_n^{-1}(\underline{Q}) - \underline{\varrho}_0) \in B] \leq P[\underline{G} \in B_\epsilon^+].$$

We may take $\epsilon \rightarrow 0$ on the right and left sides of (3.12) to conclude

$$(3.13) \quad \lim_{n \rightarrow \infty} P[n^{1/2}(\underline{S}_n^{-1}(\underline{Q}) - \underline{\varrho}_0) \in B] = P[\underline{G} \in B].$$

Finally, since (3.13) holds for every bounded open set B and since every open set in E^T is a finite or countable union of such bounded open sets, we may conclude the convergence in distribution of $n^{1/2} (\underline{S}_n^{-1}(\underline{Q}) - \underline{\theta}_0)$ to $\underline{G}$ (cf. Billingsley 1968, Theorem 2.2, p. 14).

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